BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: C Batch: 2 Practical No: 10

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Subject Name: Deep Learning

Problem Statement:

Design and implement deep learning application for audio classification that incorporates Speech to Text and Text to Speech using appropriate deep learning architecture and evaluate its performance using suitable metrics.

Algorithm:

1. Start the program and install the required libraries.
2. Import the necessary libraries and load the Whisper deep learning model.
3. Upload the input audio file in WAV, MP3, or M4A format.
4. Read the audio file and preprocess it for speech recognition.
5. Pass the audio to the Whisper model to recognize spoken words.
6. Convert the recognized speech into text and display the transcription.
7. Enter the correct reference transcript and calculate Word Error Rate (WER).
8. Pass the recognized text to the gTTS engine.
9. Generate the speech audio file, play the output, and download it.
10. Display the results, evaluate performance, and stop the program.

Code:

```
!pip -q install openai-whisper gTTS jiwer
!apt-get -qq update
!apt-get -qq install -y ffmpeg

print("Required libraries installed successfully.")
import os
import time
import whisper
from gtts import gTTS
from jiwer import wer
from google.colab import files
from IPython.display import Audio, display

# Load the Whisper deep learning model
model = whisper.load_model("base")
print("Whisper model loaded successfully.")
```

```
# Upload your speech audio file
print("Upload an audio file (WAV, MP3, or M4A):")
uploaded = files.upload()

audio_file = next(iter(uploaded.keys()))

# Perform speech recognition
start_time = time.time()
result = model.transcribe(audio_file)
stt_time = time.time() - start_time

recognized_text = result["text"].strip()
detected_language = result.get("language", "unknown")

print("\n--- Speech-to-Text Results ---")
print("Input file:", audio_file)
print("Recognized text:", recognized_text)
print("Detected language:", detected_language)
print(f"Processing time: {stt_time:.2f} seconds")
print("Original uploaded audio:")
display(Audio(filename=audio_file))

reference_text = input("Enter the correct reference transcript: ").strip()

if reference_text and recognized_text:
    error_rate = wer(reference_text.lower(), recognized_text.lower())

    print("\n--- STT Evaluation ---")
    print("Reference transcript:", reference_text)
    print("Recognized transcript:", recognized_text)
    print(f"Word Error Rate (WER): {error_rate:.4f}")
    print(f"Word Error Rate (percentage): {error_rate * 100:.2f}%")

    if error_rate == 0:
        print("Result: Exact word match.")
    elif error_rate <= 1:
        print("Result: Transcription contains word errors.")
    else:
        print("Result: High error rate; check that the reference transcript matches the audio.")
else:
    print("Reference or recognized text is empty; evaluation skipped.")
from IPython.display import display, Markdown
import textwrap

print("--- Speech-to-Text Results ---")
print("Input file:", audio_file)
print("Detected language:", detected_language)
print(f"Processing time: {stt_time:.2f} seconds")

display(Markdown(
    "### Recognized Text\n\n" +
    textwrap.fill(recognized_text, width=90)
))

if os.path.exists("/content/generated_speech.mp3"):
```

```
    print("Generated speech output:")
    display(Audio(filename="/content/generated_speech.mp3"))

    files.download("/content/generated_speech.mp3")
else:
    print("Generate the speech first by running Cell 9.")

print("=" * 45)
print("FINAL PROJECT RESULTS")
print("=" * 45)
print("Input audio:", audio_file)
print("Detected language:", detected_language)
print("Recognized text:", recognized_text)
print(f"STT processing time: {stt_time:.2f} seconds")

if reference_text and recognized_text:
    print(f"Word Error Rate: {wer(reference_text.lower(), recognized_text.lower()):.2%}")

if os.path.exists("/content/generated_speech.mp3"):
    print("TTS output: Generated successfully")
    print("Audio file: generated_speech.mp3")

print("=" * 45)

import os
from jiwer import wer

print("=" * 40)
print("EVALUATION METRICS")
print("=" * 40)

print(f"STT Processing Time: {stt_time:.2f} seconds")

if reference_text.strip() and recognized_text.strip():
    error_rate = wer(
        reference_text.lower(),
        recognized_text.lower()
    )
    print(f"Word Error Rate (WER): {error_rate * 100:.2f}%")

    if 0 <= error_rate <= 1:
        print(f"Approximate Word Accuracy: {(1 - error_rate) * 100:.2f}%")
    else:
        print("Word Accuracy: Not reported because WER exceeds 100%.")
else:
    print("WER: Reference transcript required.")

if os.path.exists("/content/generated_speech.mp3"):
    print("TTS Generation: Successful")
    print("Output Format: MP3")
else:
    print("TTS Generation: Not completed")

print("=" * 40)
```

```
start_time = time.time()
result = model.transcribe(audio_file)
stt_time = time.time() - start_time

recognized_text = result["text"].strip()
detected_language = result.get("language", "unknown")

print("\n--- Speech-to-Text Results ---")
print("Input file:", audio_file)
print("Recognized text:", recognized_text)
print("Detected language:", detected_language)
print(f"Processing time: {stt_time:.2f} seconds")
```

```
Upload an audio file (WAV, MP3, or M4A):
Choose Files  Cash_is_m...127084.mp3
Cash_is_making_a_comeback_English_Listening_B2_Ep_877-127084.mp3(audio/mpeg) - 14630696 bytes, last modified: 20/10/2026 - 100% done
Saving Cash_is_making_a_comeback_English_Listening_B2_Ep_877-127084.mp3 to Cash_is_making_a_comeback_English_Listening_B2_Ep_877-127084 (3).mp3

--- Speech-to-Text Results ---
Input file: Cash_is_making_a_comeback_English_Listening_B2_Ep_877-127084 (3).mp3
Recognized text: Remember this? This is cash. But you walk into a cafe, tap your phone and walk out with your cup of coffee. Easy, isn't it? So why in a world where we can
Detected language: en
Processing time: 32.13 seconds
```

```
--- Speech-to-Text Results ---
Input file: Cash_is_making_a_comeback_English_Listening_B2_Ep_877-127084 (3).mp3
Detected language: en
Processing time: 32.13 seconds
```

Recognized Text

Remember this? This is cash. But you walk into a cafe, tap your phone and walk out with your cup of coffee. Easy, isn't it? So why in a world where we can pay with our phones, cards or watches? Why are millions of people returning to paper money to cash? Welcome, I'm Hillary and this is a Learn Through Listening Podcast. We help you learn English by listening to interesting topics. Listening to lots of English will help you pick up lots of words naturally, mirroring how native speakers learn. So in this episode, lots of vocabulary around cash, C-A-S-H and payments and money and bank transactions. So this is really useful material. Hello, I'm Hillary and you're listening to Adept English. We will help you to speak English fluently. Or you have to do is listen. So start listening now and find out how it works. And an interesting topic. We are looking today at the possible effects of a cashless society. That means a society with no physical money. And right at the end, I'll talk about why I think this is a bad thing. Why I want physical cash to stay. And it's not a reason that many other people are talking about. But I think we all feel it quite strongly. Listen right to the end to find out my thoughts. So think about the last time you bought something. It's quite possible you didn't end over an actual bank note. Like this 10 pound note. We live in an era where tapping a plastic card or even just your smartphone to pay for something is absolutely normal. It's very convenient. I can pop out to my local waiter of supermarket with just my car keys and my phone. Great. And this means that payments in shops are much quicker. There's less need for actual staff to take your payment, actual people at the checkout. And you just pay at a machine. And there is part of the problem. There's no need to speak to anyone. You just pay at the machine. This can mean staff lose their jobs, especially in supermarkets. And everyone just does less interaction with other people. When everything is just a digital payment and a tap, we perhaps lose something of that human connection. But I also get it. It's cheaper for shops to have machines rather than people. And it would be really hard to stop progress on this one. And with many machines, you can actually pay by cash anyway. So that's more a way of getting around needing people. But for many people, the loss of cash would mean much more than missing a friendly chat. It creates a problem for people who are vulnerable. If you are vulnerable, that's V-U-L-N-E-R-A-B-L-E. It means you're at risk or in need of protection. Older people, those on very low incomes or people with certain disabilities, can be vulnerable in our society. If a country went completely digital, there are people who would be left behind. Professor Alistair Milner from Loughborough University in the UK and Professor Marcus Sakaryadis from Manchester University published a really interesting report in October 2025. They warned that the UK risks leaving people totally excluded, totally left out from the economy. The digital revolution is great if you have a smartphone and you're part of it. But it can leave a lot of people behind. The report described, for example, how digital wallets. That's things like Apple Pay and Google Pay. How they were only 9% of transactions in 2019. But in 2023, they were nearly a third. So a massive rise in these types of payments. A digital wallet? Well, a wallet, W-A-L-L-E-T, it's often made of leather and it's where you keep your bank cards and other cards. But we're talking here about a digital wallet. So

```
print("=" * 40)
```

```
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EVALUATION METRICS
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STT Processing Time: 32.13 seconds
Word Error Rate (WER): 8200.00%
Word Accuracy: Not reported because WER exceeds 100%
TTS Generation: Successful
Output Format: MP3
----------------------------------------
```